

CLASS 07 / OCT 6 / U04

ARRAYS

HEX · RADIAL · CURVE

P3b SHEET 2 — STAGGER IT, SPIN IT, BEND IT

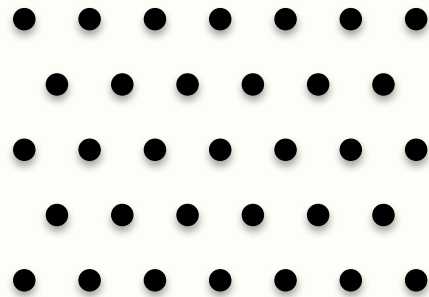
NOTHING DUE

STUDIO-REVIEW WEEK — NOTHING DUE TODAY / P3b DUE WEEK 8 — OCT 13 / 7%

SHEET 2: THREE MORE RULES

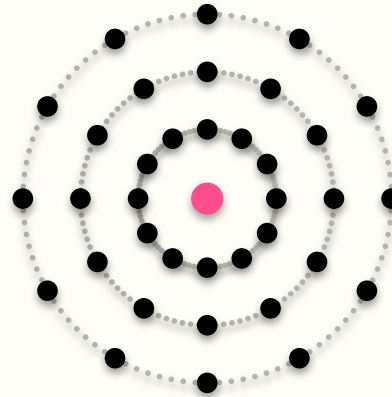
Same module. Three new rules for turning an index into a position.

HEXAGONAL



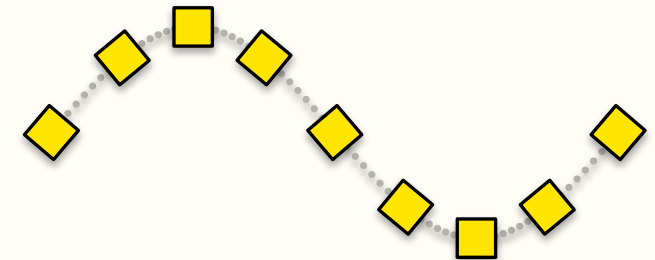
stagger alternate rows

RADIAL



radius, count, angular step

CURVE



resample, place, align

Sheet 1 counted along axes. Sheet 2 keeps the counter and changes the rule that turns it into a position.

HEXAGONAL IS A GRID WITH AN ODD/EVEN TEST

Stagger alternate rows by half the spacing. Set the row spacing on its own.

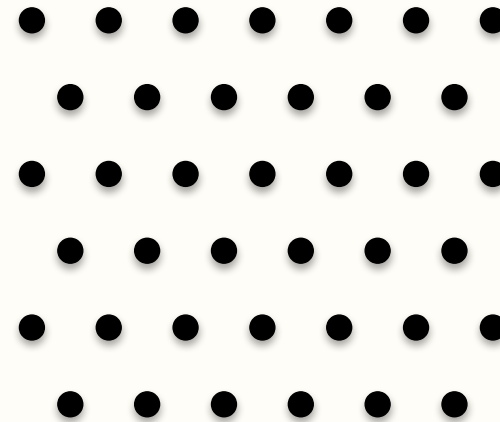
THE TEST

```
offset = (j mod 2 == 0) ? 0 : dx / 2
```

Modulo asks: is this row odd or even? Compare turns that into True or False, and Switch picks the offset. The same two nodes as last week.

Row spacing is not dx. For a true hex field it is $dx \times \sqrt{3} / 2$. Use dx for both and you get a squashed grid.

WHAT IT LOOKS LIKE



A BRICK BOND IS A STAGGERED ARRAY. SO IS A HONEYCOMB. SO IS HALF THE PAVING YOU WALK ON.

RADIAL IS POLAR ARITHMETIC

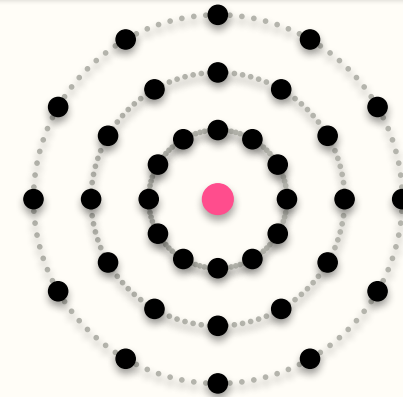
Radius, count, angular step. The index drives the angle.

THE RULE

```
step  = 360 / count  
 $\theta[i]$  =  $i \times \text{step}$   
 $p[i]$  =  $c + r \cdot (\cos \theta, \sin \theta)$ 
```

The rotation arithmetic from Week 2 — where x and y have to talk to each other — now generating positions instead of moving a solid.

THREE KNOBS



Change the count and the step changes with it. Change the radius and the ring opens up.

IF YOUR LAST COPY LANDS ON TOP OF YOUR FIRST, YOU DIVIDED BY THE WRONG NUMBER.

AN ARRAY ALONG A CURVE

The curve supplies the positions. Its tangent supplies the direction.

THREE STEPS

01

RESAMPLE

Divide the curve into a count of evenly spaced points.

02

PLACE

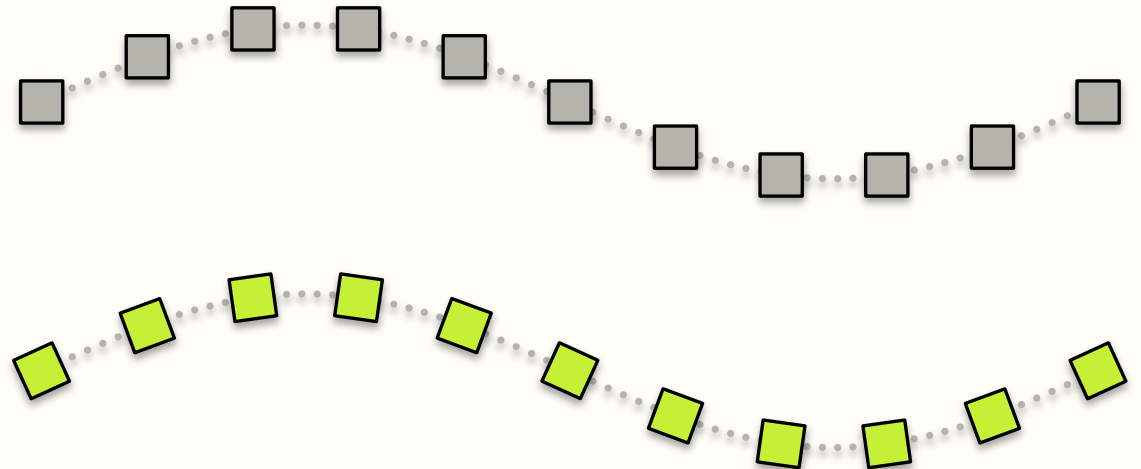
Put one copy of the module on each point.

03

ALIGN

Rotate each copy to the curve's tangent at that point.

NOT ALIGNED · ALIGNED



WITHOUT THE TANGENT, EVERY COPY FACES THE SAME WAY. WITH IT, THE ROW FOLLOWS THE CURVE.

DRIVE THE CURVE WITH SINE

Three words describe the whole curve. Sheet 2 asks you to label all three.

THREE PARAMETERS

AMPLITUDE

how far it swings

FREQUENCY

how many swings along the curve

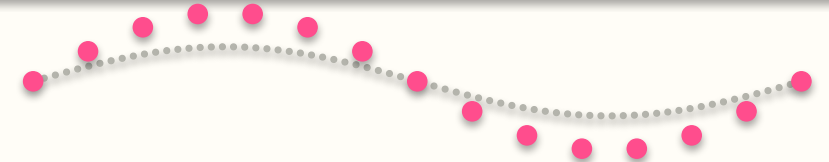
PHASE

where in the swing it starts

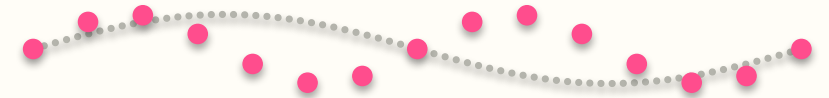
$$y = \text{amplitude} \times \sin(x \times \text{frequency} + \text{phase})$$

CHANGE ONE AT A TIME

AMPLITUDE $\times 2$



FREQUENCY $\times 2$



PHASE $+ 90^\circ$



Grey: the base curve. Pink: one value changed.

SET EACH VALUE ON PURPOSE AND WRITE IT ON THE SHEET. A SINE NOBODY CAN READ IS DECORATION.

TWO WAYS TO REPEAT

Worth being precise about, because they look alike and are not.

EXPLICIT ITERATION

A Repeat Zone runs a body once per step, in order, carrying a value forward.

Use it when step n depends on step $n - 1$ — and when you want to watch the loop happen.

PER-ELEMENT EVALUATION

A field is evaluated for every element at once. There is no order and no carry.

It is usually faster — but nothing on screen tells you a loop happened, which is why we built the explicit one first. A field is not a Repeat Zone.

NEXT WEEK'S ATTRACTORS ARE MEASURED PER ELEMENT. KNOW WHICH KIND OF REPETITION YOU ARE USING.

SHEET 2, STEP BY STEP

What we build together today: hexagonal, radial, curve.

01

Week 7 · Hexagonal: stagger alternate rows; explain the odd/even test and row spacing.

02

Week 7 · Radial: place copies from radius, count and angular step.

03

Week 7 · Curve: resample a curve and place copies along it, aligned to its tangent. Drive a curve with sine; explain amplitude, frequency and phase.

THE WHOLE UNIT, WRITTEN OUT

Last week you could read the loops. Today you can read all of it.

```
points = []

for j in range(rows):                # nested loop = grid
    offset = spacing / 2 if j % 2 else 0  # odd/even stagger
    for i in range(cols):
        z = amplitude * sin(i * frequency + phase)
        if 3 <= i <= 5:                # Compare + Switch
            z = z + lift
        points.append([i * spacing + offset, j * row_gap, z])
```

WHAT IT MAPS ONTO

list	modulo
index	sine
count	amplitude
loop	frequency
nested loop	phase
conditional	

j % 2 IS THE ODD/EVEN TEST · if IS COMPARE AND SWITCH · sin IS THE MATH NODE

REMINDER — P3b REQUIREMENTS

Visual quality 35% · computational understanding 25% · technical execution 20% · experimentation 10% · requirements and identity 10%.

01

Array your P3a module across exactly two 17×11 inch sheets, submitted together as one assignment.

02

Sheet 1 — 1D, 2D and 3D arrays: a linear array, a nested grid and an XYZ cube array. Label count, spacing and index order; show the predicted total for each.

03

Sheet 2 — hexagonal, radial and curve arrays: a staggered hexagonal grid, a radial array (radius, count, angular step) and an array along a curve, including one sine-driven curve with amplitude, frequency and phase labelled.

04

Keep the module unchanged; vary only its placement. Expose count and spacing or radius in the graph.

05

Build repetition from basic nodes and begin with a count you can inspect; there is no minimum element count.

06

Include one conditional selection on either sheet. State its rule (for example, indices 3–5 lift by 0.5) and show the array before and after it applies.

NOW — SHEET 2

Nothing is due today; this is a studio-review week. Use the class to build sheet 2.

✓

Hexagonal array working, the odd/even test visible and row spacing set

✓

Radial array working from radius, count and angular step

✓

Your module placed along a resampled curve, aligned to its tangent

✓

A sine-driven curve, with amplitude, frequency and phase written down

✓

Sheet 1 reviewed — labels in place, and one conditional selection with its rule stated

TWO SHEETS, ONE MODULE, SIX ARRAYS. NOTHING ELSE CHANGES.

BEFORE NEXT CLASS

NEXT

Week 8: attractors — distance, a 2D grayscale field, remap, clamp and falloff; point, multi-point and curve attractors.

DUE

P3b is due next class — both sheets, submitted together as one assignment. DUE WEEK 8 — Oct 13 / 7%.

FINISH

Sheet 2: hexagonal, radial and curve arrays, the sine values labelled. Check sheet 1 against the requirements again.

PRINT

P2b — the printed object and its process sheet are due at the mid-semester deadline, WEEK 10 — Oct 27. If it is not queued yet, queue it now.